Analysis of the Islamic Scholarly Network in the Hadith *Sanad* Using Knowledge Graph and Social Network Analysis with an Interactive Dashboard Visualization

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***Abstract*—The hadith *sanad* is an Islamic scholarly network formed by chains of teacher–student relationships among narrators across generations, a relational structure that lends itself naturally to directed-graph modeling. Prior studies have applied Knowledge Graph (KG) and Social Network Analysis (SNA) methods to *sanad* networks, but their outputs are static, rendered at full scale in generic graph terminology, and operable only through desktop software such as Gephi and Cytoscape, so hadith students and researchers—the very audience that routinely traces narrator standing—must still work through *rijal* literature sequentially by hand. This paper analyzes the Islamic scholarly network of the *sanad* as a KG with SNA and presents the analysis through a publicly accessible interactive web dashboard. From the Sanadset 650K corpus (650,986 hadith entries from 926 classical Arabic books), 487,451 cleaned *sanad* chains were modeled as a weighted directed graph of 177,047 narrator nodes and 776,798 narration edges. In-degree, out-degree, and eigenvector centrality were computed on the full graph; closeness and betweenness were computed on a 14,695-node strongest-weight subgraph. Abu Hurairah ranks first in four of five metrics while Sufyan leads in out-degree, a role split that matches classical biographical assessments. The residual risk of identity merging introduced by kinship-term resolution was quantified rather than assumed, and is concentrated in 9.85% of the 102,100 resolution events. Building and analyzing the full graph takes about 13 s for loading, construction, and the degree- and eigenvector-based metrics, with a 701.4 MB peak RSS. The dashboard, built with React over a Supabase (PostgreSQL) backend and an HTML5 Canvas force-directed graph, passed 14 of 14 black-box scenarios, sustained 57–61 FPS during graph interaction, and scored 71.23 on an adapted seven-item System Usability Scale with 18 respondents.**

***Index Terms*—knowledge graph, hadith sanad, social network analysis, narrator centrality, network visualization, interactive dashboard**

I. Introduction

Hadith, the recorded sayings, actions, and tacit approvals of the Prophet Muhammad, is the second source of Islamic law after the Qur'an [1], [2]. Structurally, a hadith consists of two components: the *matn* (the text itself) and the *sanad*, the continuous chain of narrators who transmitted the text across generations. The *sanad* forms an Islamic scholarly network built from teacher–student relationships, in which each narrator is a link in the intergenerational transfer of religious knowledge [3]. A narrator is conventionally identified by tracing the list of their teachers and students together with their generation (*tabaqat*) [4], so a *sanad* is naturally represented as a directed graph with narrators as nodes and teacher–student relations as edges [5].

Mapping this network accurately is not straightforward. The source material is classical Arabic text with high orthographic variance in narrator names—presence or absence of diacritics, alternate Unicode letterforms, contextual kinship terms, and the *kunyah* forms *Abu*, *Abi*, and *Aba*—so without careful handling one narrator is recorded as several distinct entities. That complexity is compounded by scale: hundreds of thousands of transmission entries drawn from hundreds of books [6], which makes manual tracing impractical.

Not all narrators play an equal role. Some occupy strategic positions as convergence points traversed by many *sanad* paths, known as *common links* [7]. Classical scholarship identifies them manually through *isnad*-bundle analysis, but at the scale of hundreds of thousands of narrators this is inadequate. Simple text search is no shortcut either: it returns a single occurrence count without capturing a narrator's structural position, a limitation consistent with the broader challenge of large-scale information retrieval, which demands richer data representations than string matching [8].

Over the past decade this need has been addressed by quantitative studies of the narrator network: reconstructing temporal information from transmission structure [9], representing the network as a Knowledge Graph (KG) [5], and applying centrality metrics together with exploratory visualization prototypes [10]. These studies generally rely on desktop analysis tools such as Gephi and Cytoscape [5], and their outputs are static images or whole-graph files that are dense and hard to trace at the scale of hundreds of thousands of narrators. They also remain expressed in generic graph vocabulary (node, edge, centrality score) without translation into roles meaningful within hadith studies, such as teacher, student, or *common link*. As a result, a hadith researcher wishing to trace one narrator's relations across generations must still do so sequentially and manually through *rijal* literature.

The KG representation nonetheless offers a foundation that can be developed into a more interactive and accessible system. A KG captures entities and their relations semantically and has been widely adopted across domains such as education and tourism to extract insight from large-scale data [11]. For hadith studies, it fits the narration network directly, with narrators as entities and teacher–student relations as edges [5]. Turning that network into explorable knowledge requires an interactive web front-end for three reasons: *visualization scalability*, so a network of hundreds of thousands of narrators is presented as a representative, progressively traceable subset rather than one dense static image [12]; *semantic accessibility*, so centrality values and graph structure are translated into terms familiar in hadith studies; and *relational navigation*, so a narrator's teachers and students can be traversed by click-through interaction instead of sequential manual lookup. Meeting these requirements is not merely a matter of choosing a front-end framework, since interactive graph performance depends heavily on the rendering technology—the choice between ready-made libraries and native browser primitives (SVG, Canvas, WebGL) determines whether a visualization stays responsive at thousands of nodes or stalls at hundreds [13].

This paper analyzes the Islamic scholarly network of the hadith *sanad* using a KG and SNA over the publicly available Sanadset 650K dataset [6], and presents the results through an interactive dashboard. The contributions are fourfold: (i) construction of a large-scale weighted directed KG of the *sanad* network through systematic preprocessing, together with a quantitative evaluation of the residual identity ambiguity that rule-based kinship resolution introduces; (ii) quantitative identification of narrator roles using five SNA centrality metrics, interpreted against classical biographical scholarship; (iii) an interactive, publicly deployed dashboard designed around the stated needs of hadith students and researchers; and (iv) empirical measurement of the computational cost of building and analyzing the graph and of the dashboard's runtime behavior, figures that prior *sanad* network studies do not report.

II. Related Work

Computational research on the hadith *sanad* has progressed from dataset construction toward graph-based modeling and SNA. Mghari *et al.* built Sanadset 650K, a corpus of 650,986 entries from 926 classical Arabic books, in which each entry separates the full hadith text, *matn*, narrator list, and *sanad* ordering [6]. Its scale and coverage make it the most representative source for *sanad* transmission research, though it is provided as raw data without graph visualization or network analysis. The same group later proposed Narrator2Vec, a Skip-Gram Word2Vec adaptation that learns vector representations of narrator names and predicts missing narrators in broken chains [10]; it does not, however, produce per-narrator quantitative scores that users can compare.

Farooqi *et al.* introduced Multi-IsnadSet, representing the Sahih Muslim transmission network as a directed graph of 2,092 nodes and 77,797 edges, built and analyzed with NetworkX, Neo4j, Gephi, and Cytoscape [5]. They demonstrated that a *sanad* network can be modeled as a KG, and explicitly noted that SNA metrics such as degree, betweenness, and eigenvector centrality *could* be applied to identify influential narrators—an analysis they left as future work. Shuaib *et al.* applied trophic analysis to a 49,309-narrator hadith network and showed that the network is hierarchical and sequential, recovering temporal information from structure [9], yet without centrality metrics or an interactive interface. Mahmoud *et al.* proposed NarratorsKG, the first KG dedicated to hadith narrators (79,256 nodes, 197,183 edges), used for narrator disambiguation with graph-query embeddings and AraBERT [14]; it is an internal NLP resource rather than an end-user system. Hendawi *et al.* used machine learning to verify *sanad* continuity from narrator biographical data [15], reinforcing the value of structured narrator data.

A parallel line of work visualizes narrator chains as graphs. Saraçoğlu converted 2,258 *isnad* chains into source–target pairs and visualized them with Pajek, Gephi, and Isnalyser [16]. HadithNet automated narrator-name identification and chain generation on a web system using Django, MySQL, and Graphviz, reaching 99.37% name precision but struggling with complex branching chains [17]. Ahmad *et al.* showed that transforming *isnad* into directed graphs enables centrality, clustering, and bridge-narrator analysis [18], and the KASHAF system offered interactive Sankey-flow visualization over 60,000 hadith [19], though focused on retrieval and *takhreej* classification rather than narrator-centrality exploration.

Across these studies three gaps persist. Dataset-oriented works yield large corpora without an exploration interface; analytical works produce models operable only through desktop software or technical expertise; and none reports the computational cost of building or querying the graph systems they describe, so no efficiency baseline exists for *sanad* networks at scale. This work addresses all three.

III. Materials and Methods

The research was conducted in four stages: preprocessing of the Sanadset 650K dataset, KG construction from teacher–student relations, network analysis using five centrality metrics, and development of an interactive web dashboard. The analytical pipeline was implemented in Python 3.13 with NetworkX 3.6, and the front-end in React 18 with a Supabase (PostgreSQL) backend.

TABLE I
DATA REDUCTION ACROSS PREPROCESSING STAGES

Stage	Process	Rows	Removed
1	Raw data	650,986	—
2	Remove chainless rows (No SANAD)	491,428	159,558
3	Remove single-*Abu* rows	488,195	3,233
4	Remove kinship-led rows	487,451	744
	Final	**487,451**	**163,535**

A. Dataset

The primary input is `sanadset.csv` from Sanadset 650K [6], a static CSV of 650,986 hadith records drawn from 926 classical Arabic books. Each record contains the source book, hadith number, hadith text, *matn*, narrator list, and *sanad* length. Only the *sanad* column is used, as it carries the narrator chain required for graph construction and centrality computation.

B. Data Preprocessing

Because narrator names in classical texts are written inconsistently, a single narrator may otherwise be split into several nodes, distorting the graph and its centrality values. Seven preprocessing steps were applied:

1) **Removal of chainless rows.** Rows marked `No SANAD` carry no narration relation and were removed, eliminating 159,558 rows.
2) **Diacritic (*harakat*) removal.** Arabic vocalization marks were stripped via regular expressions so that the same name written with or without diacritics maps to one entity.
3) **Arabic-letter normalization.** Unicode variants of *alif*, *ya*/*alif maqsura*, and *ta marbuta* were unified to canonical forms.
4) **Kinship-term resolution.** Contextual terms meaning "his father," "his mother," or "his grandfather" were replaced by explicit constructions referencing the preceding narrator, preventing generic merge nodes.
5) **Transmission-term removal.** Narration formulas (e.g., *haddathana*, *akhbarana*, *'an*) were removed so they do not form spurious nodes; 185 chains still containing such formulas were cleaned in a second pass.
6) ***Abu* variant normalization.** Grammatical forms *Abu*/*Aba*/*Abi* were unified to the nominative *Abu*.
7) **Non-Arabic character validation.** Characters outside the Arabic Unicode range were removed; none remained after the prior steps.

Additional removal of ambiguous rows (single *Abu* elements and rows beginning with kinship terms) left 487,451 clean *sanad* chains for graph construction, as summarized in Table I.

C. Knowledge Graph Construction

The *sanad* network is modeled as a KG of structured triples (Student, narrates-from, Teacher). Each chain of n narrators is decomposed into $n-1$ triples, one per adjacent pair. Because transmission flows directionally from student to teacher, the network is built as a directed graph using `nx.DiGraph()`, with each student node as the source and each teacher node as the target. Self-loops are removed.

Each edge carries two weight attributes. *Weight* is the frequency with which a teacher–student pair occurs across the corpus; a higher weight indicates a stronger, more frequently attested transmission. *Inverse Weight*, the reciprocal $1/\text{Weight}$, serves as the edge distance for path-based centrality: frequent relations imply closer narrators (shorter distance), while rare relations imply greater distance. The final graph comprises **177,047 narrator nodes** and **776,798 narration edges**. Narrator names in Arabic script were additionally transliterated to Latin script for readability.

D. Evaluating Kinship-Resolution Ambiguity

Kinship resolution is valid locally within one chain, because a kinship term necessarily refers to the immediately preceding narrator. When identically named resolved nodes (e.g., "Abu Muhammad") arise from different chains, however, string equality does not guarantee identity, so several narrators may be merged into one node. Rather than assume this risk away, it was measured. The preprocessing stage was re-executed under instrumentation that logs every resolution event with its source book, hadith number, anchor name, and resulting node; replicated node names matched the nodes of the production graph at 99.59% (14,173 of 14,231).

The evaluation proceeds on two levels. First, an *absorption rate* groups all resolution events by resulting node. Because high frequency alone may simply reflect a canonical *isnad* quoted identically across many books, each anchor was classified by specificity into *bare* anchors (a single-word name with no distinguishing *nasab*, e.g. *Muhammad*), which carry high merge risk, and *qualified* anchors (a name with *nasab*, e.g. *Hisyam bin Urwah*), which in practice still denote one individual. Second, a *degree comparison* contrasts total degree (in-degree plus out-degree) across direct nodes, qualified-anchor resolved nodes, and bare-anchor resolved nodes, using the Mann-Whitney U test: a node merging the relations of several individuals should show anomalously high degree. Manual inspection of the highest-frequency, highest-degree nodes complements both levels.

E. Centrality Metrics

Five SNA centrality metrics [20] were computed for every node, each capturing a distinct narrator role.

In-degree centrality measures incoming edges, i.e., how many students narrate from a narrator (role as a source/teacher):

$$C_D^{in}(v) = \frac{\deg^{in}(v)}{n-1}. \tag{1}$$

Out-degree centrality measures outgoing edges, i.e., how many teachers a narrator received from (role as an active student):

$$C_D^{out}(v) = \frac{\deg^{out}(v)}{n-1}. \tag{2}$$

Eigenvector centrality measures influence by connection quality—being linked to other influential narrators:

$$C_E(v) = \frac{1}{\lambda} \sum_{u \in N(v)} A_{vu}\, C_E(u), \tag{3}$$

where λ is the largest eigenvalue of the adjacency matrix A and $N(v)$ is the neighbor set of v.

Closeness centrality measures proximity to all other nodes by shortest path, using *Inverse Weight* as distance:

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)}. \tag{4}$$

Betweenness centrality measures how often a node lies on shortest paths between other pairs, marking bridge narrators:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{5}$$

where σ_{st} is the number of shortest s–t paths and $\sigma_{st}(v)$ the number passing through v.

In-degree, out-degree, and eigenvector centrality are computationally light and were computed on the full graph. Closeness and betweenness require all-pairs shortest-path traversal, which is intractable on the full graph; they were computed on a representative subgraph of the 50,000 strongest-weight edges (14,695 nodes), preserving the principal transmission backbone.

F. Computational Benchmarking

To ground that coverage decision quantitatively, a Python benchmark script instrumented every pipeline stage with wall-clock timing and process memory (Resident Set Size), covering CSV loading, graph construction (`from_pandas_edgelist`), self-loop removal, weighted degree, and the five centrality metrics on the full 177,047-node graph. Closeness was computed exactly but only over 100 randomly selected nodes (seed 20260721), and betweenness using NetworkX's official sampling parameter ($k = 100$ source nodes), so that the cost of path-based metrics could be measured within a reasonable time budget. The benchmark also records structural properties—density, mean degree, weakly and strongly connected components, and an approximate diameter via `nx.approximation.diameter` on the largest component. All measurements were taken on one environment (Table II).

TABLE II
BENCHMARK ENVIRONMENT

Component	Specification
Operating system	Windows 11 (build 10.0.26200)
Processor	AMD64, 6 cores / 12 threads
Memory (RAM)	6.29 GB
Python	3.13.13
NetworkX	3.6.1
pandas	3.0.2

G. System Architecture and Dashboard

The primary target users are students and researchers of hadith studies, chosen because they routinely trace *sanad* and assess narrator standing, because the concepts measured by centrality have counterparts they already know (such as the *common link* [7]), and because they typically lack the technical background to operate desktop network software [5]—making them the group most affected by the accessibility gap. Six user needs were derived and mapped onto dashboard features: (K1) gauging a narrator's position and influence quantitatively; (K2) tracing the teachers and students connected to a narrator; (K3) comparing roles across narrators; (K4) obtaining an overview of the most prominent narrators without traversing the graph node by node; (K5) inspecting relation and centrality data textually for citation; and (K6) accessing all of this without installing software.

The system follows a three-tier architecture (Fig. 1) separating heavy computation from data presentation. The *data processing layer* runs offline in Python, performing preprocessing, KG construction, and centrality computation, then exporting two CSV files: `sentralitas.csv` (per-narrator centrality scores) and `edge_slim.csv` (weighted relations). The *data storage layer* is Supabase (PostgreSQL), holding a CENTRALITY table (nodes) and an EDGE table (relations) linked by narrator ID, queryable via REST API. The *presentation layer* is a read-only Single Page Application built with React, TypeScript, and Vite, styled with TailwindCSS, that retrieves data through `@supabase/supabase-js` and renders four pages. Statistical charts use Recharts, while the interactive network graph is rendered directly on an HTML5 Canvas with a custom force-directed layout. Placing all computation in the processing layer ensures centrality is computed once on the backend rather than on the client, keeping the dashboard lightweight and responsive.

The force-directed layout combines three forces per animation frame: a Coulomb-like *repulsion* $F_r = k_r/d^2$ ($k_r = 2000$) applied between node pairs closer than 320 px, a Hooke-like *spring* force $F_s = (d - L_0)k_s$ ($L_0 = 200$ px, $k_s = 0.003$) along edges, and a *radial constraint* $F_c = (r - r_{target})k_c$ ($k_c = 0.014$) that pulls each node toward a concentric ring keyed to its centrality rank, so the most central narrators settle near the graph center. Velocities are damped by a factor of 0.78 per frame. Centrality is further encoded through node size (28, 20, 14, 10, and 7 px by rank group) and color, and edges are drawn as arrows pointing from student to teacher.

H. Testing and Evaluation

Four complementary evaluations were performed. *Black-box testing* covered 14 scenarios spanning navigation, data loading, graph rendering, node selection, metric filtering, search, zoom, drag and pan, auto-fit reset, the statistics page, both tables, pagination, and sorting.

Non-functional testing follows the *Performance Efficiency* characteristic of ISO/IEC 25010 as applied to web information

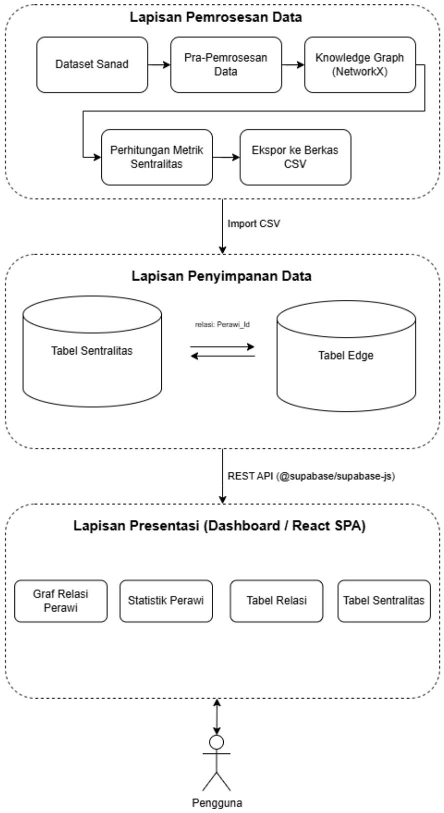

Fig. 1. Three-tier system architecture separating offline analysis, centralized storage, and the interactive presentation layer.

systems by Rajata *et al.* [21], covering *Time Behaviour*, *Resource Utilization*, and *Capacity*. Frame rate was sampled once per second through a Playwright-injected frame counter during zoom, pan, and node drag, and judged against the common thresholds of 60 FPS for smooth animation and 30 FPS for perceptible jank [13]. Browser memory was measured as the active JavaScript heap via the Chrome DevTools Protocol (`Performance.getMetrics`) with forced garbage collection before each reading, at node counts N increasing toward full network scale. API load testing used k6 against the Supabase REST endpoints with a ramping virtual-user (VU) profile, with success criteria of p(95) latency below 1,000 ms and a request failure rate below 1%.

Usability evaluation used an online questionnaire adapting seven of the ten standard System Usability Scale (SUS) statements [22], omitting the three items on cumbersomeness, user confidence, and prior learning. Because the item count differs, the standard multiplier of 2.5 does not apply: per-item contributions were computed as usual (positive items as response minus 1, negative items as 5 minus response), and the resulting 0–28 total was rescaled to 0–100 by a factor of $100/28$. The result is therefore an *adapted* SUS score, not a standard one. Eighteen respondents completed the questionnaire, which also collected demographics and three open-ended questions.

TABLE III
STRUCTURAL CHARACTERISTICS OF THE SANAD GRAPH

Parameter	Value
Density	2.478×10^{-5}
Mean node degree	8.775
Weakly connected components	51
Largest weakly connected component	176,891 (99.91%)
Strongly connected components	47,300
Largest strongly connected component	129,721
Approximate diameter (largest WCC)	28

IV. Results and Discussion

A. Graph Construction and Structure

Normalization yielded 177,289 unique narrator names; of these, 177,047 participate in at least one narration relation and form the graph nodes, the remaining 242 being isolated names (single-narrator chains or names dropped during self-loop and empty-name cleaning). The resulting directed graph contains 177,047 nodes and 776,798 edges. The strongest relation is Ibnu Umar $\leftarrow$ Nafi' with a weight of 11,346, indicating that Nafi' narrated from Ibnu Umar that many times in the corpus; such high-weight pairs form a dominant transmission backbone that recurs across many chains.

Table III reports the structural profile. The density of 2.478×10^{-5} marks the graph as sparse: each narrator connects directly to 8.775 others on average out of 177,046 possible partners, characteristic of large real-world social networks. A single weakly connected component holds 99.91% of all nodes, so the *sanad* network is one interconnected structure rather than isolated islands of transmission. Strongly connected components are far more numerous (47,300), consistent with the one-way nature of narration: mutual reachability requires a directed teacher–student path in the same direction, which is far rarer. The approximate diameter of 28 on the largest component indicates that the longest path between any two narrators remains short relative to a network of this size, consistent with small-world behavior.

B. Computational Performance of Graph Construction and Analysis

Table IV reports the cost of each pipeline stage on the full graph. Data loading, graph construction, and the four degree- and eigenvector-based metrics together take roughly 13 s at negligible memory cost. Closeness and betweenness are dramatically more expensive—about 9.4 and 3.8 minutes respectively—*even under sampling* at $k = 100$ rather than exact computation over all 177,047 nodes. Extrapolating from the sampled timings, exact betweenness on the full graph would require more than 50 hours, which quantitatively confirms the methodological decision to restrict both path-based metrics to the 14,695-node representative subgraph. Peak process memory reached only 701.4 MB on a 6.29 GB

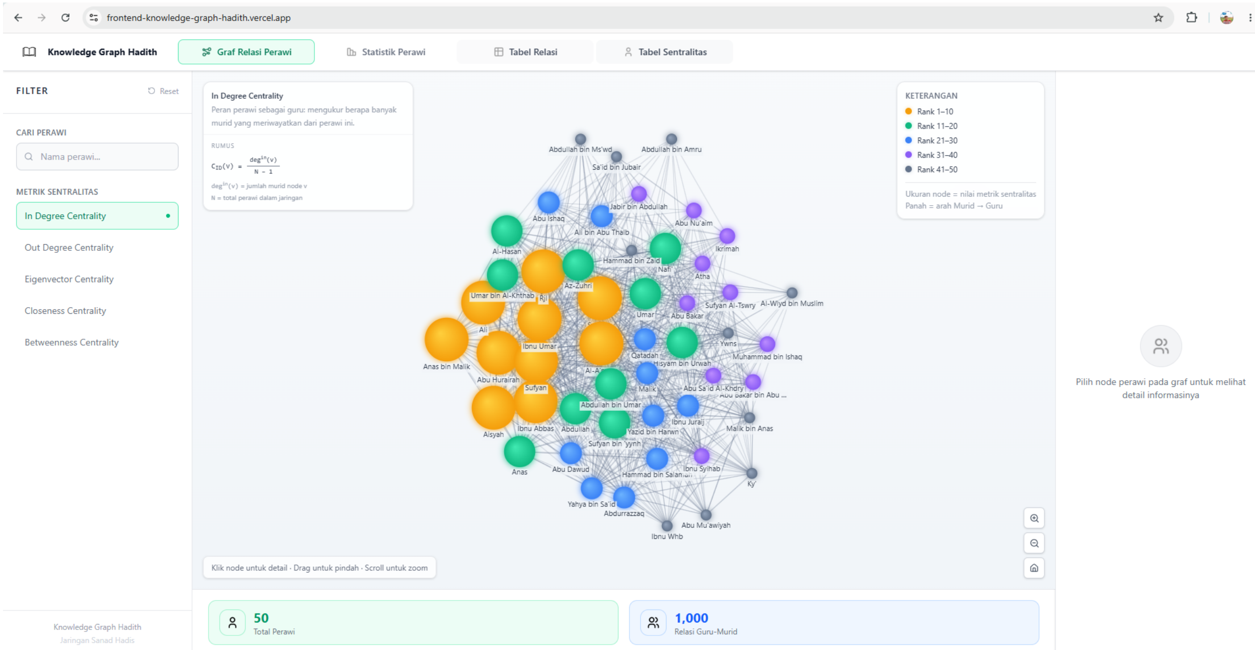

Fig. 2. Narrator Relation Graph page of the dashboard. Node size and color encode centrality rank (In-degree selected); arrows point from student to teacher. The left panel filters by metric and search; the right panel shows per-narrator detail when a node is selected.

TABLE IV
COMPUTATIONAL COST OF KG CONSTRUCTION AND ANALYSIS

Stage	Time	Δ RSS
CSV loading	2.07 s	+172.6 MB
Graph construction	4.09 s	+300.9 MB
Self-loop removal	0.06 s	≈0 MB
Weighted degree	0.86 s	+1.8 MB
Degree centrality	0.19 s	+1.4 MB
In-degree centrality	0.16 s	≈0 MB
Out-degree centrality	0.15 s	≈0 MB
Eigenvector centrality	5.10 s	+7.3 MB
Closeness centrality (sampled, k=100)	562.83 s	—
Betweenness centrality (sampled, k=100)	228.57 s	—
Peak process memory (peak RSS)		**701.4 MB**

machine, showing that a 177,047-node KG can be built and analyzed on modest hardware; execution time, not memory, is the binding constraint.

C. Residual Ambiguity of Kinship Resolution

Kinship resolution fired 102,100 times across the 487,451 chains, dominated by *Abihi* ("his father," 93,066 events, 91.1%), followed by *Jaddihi* (7,287, 7.1%), *Ummihi* (1,004, 1.0%), *Abiha* (425, 0.4%), *Jaddatihi* (234, 0.2%), and *Abaihi* (84, 0.1%). These events produced 14,231 string-unique nodes, an average of 7.17 events per node. That average alone does not indicate identity collision, since a canonical *isnad* quoted identically across many books produces the same repetition. After classification by anchor specificity, only 623 of 14,231

TABLE V
GRAPH DEGREE BY NODE GROUP

Group	Nodes	Mean deg.	Median deg.
Direct	162,874	8.81	2
Resolved (qualified anchor)	13,552	5.66	2
Resolved (bare anchor)	621	66.71	4

nodes (4.4%) carry bare anchors—yet this small group accounts for 10,059 of 102,100 events (9.85%), a disproportionate share. The highest-frequency collision candidates include *Abu Salim*, *Abu Mu'tamir*, *Abu Rajul*, *Abu Sufyan*, *Umm Al-Hasan*, *Abu Ibrahim*, *Abu Jaddi*, and *Abu Ahmad*.

The degree comparison in Table V makes the effect visible on the graph itself. A naive comparison of all resolved nodes against direct nodes, without distinguishing anchor specificity, shows no significant difference (mean 8.33 vs. 8.81; median 2 vs. 2; Mann-Whitney U $p = 0.275$), which would have suggested that kinship resolution does not inflate degree at all. Splitting by anchor specificity reverses that reading: bare-anchor nodes average a degree of 66.71, far above both other groups, with very strong statistical significance ($p \approx 4 \times 10^{-45}$ against direct nodes and $p \approx 7 \times 10^{-47}$ against qualified-anchor nodes), while qualified-anchor resolved nodes are in fact slightly *lower* in degree than direct nodes ($p = 0.99$).

Manual inspection agrees. *Abu Salim* (1,492 events, degree 625) is dominated by "*Salim 'an abihi*" chains through Az-Zuhri, most likely Salim bin Abdullah bin Umar, consistently leading to Umar bin Khattab—but a minority tail leads to Hafshah, Zaid bin Tsabit, and a different *Salim*, indicating

TABLE VI
TOP NARRATORS BY CENTRALITY METRIC

Rank	Metric	Narrator	Score
1	In-degree	Abu Hurairah	0.0199
2	In-degree	Ibnu Abbas	0.0150
3	In-degree	Aisyah	0.0132
1	Out-degree	Sufyan	0.0135
2	Out-degree	Abu Abdullah Al-Hafizh	0.0127
3	Out-degree	Syu'bah	0.0124
1	Eigenvector	Abu Hurairah	0.5195
2	Eigenvector	Aisyah	0.4836
3	Eigenvector	Ibnu Umar	0.3821
1	Closeness	Abu Hurairah	1.0000
2	Closeness	Aisyah	0.9965
3	Closeness	Ibnu Abbas	0.9864
1	Betweenness	Abu Hurairah	0.0500
2	Betweenness	Syu'bah	0.0445
3	Betweenness	Sufyan	0.0384

partial contamination. *Abu Mu'tamir* (234 events, degree 126) shows a far more heterogeneous successor distribution across roughly 80 books, a stronger collision signal. By contrast *Abu Hisyam bin Urwah* (6,562 events), a qualified anchor, consistently denotes one individual despite being among the most frequent, confirming that frequency alone is not a reliable indicator.

Taken together, the risk is measurable and concentrated rather than pervasive: 9.85% of resolution events carry high merge risk, while the remaining 90.15% behave statistically like direct-name nodes. The 4.4% bare-anchor share of resolved nodes can therefore be reported as an upper bound on identity-collision risk, in place of an unevidenced claim that all resolved names are ambiguous.

D. Centrality Analysis

Table VI lists the top narrators for each metric. Abu Hurairah ranks first in four of five metrics—in-degree, eigenvector, closeness, and betweenness—identifying him as the most central and most influential narrator in the network. Sufyan ranks first in out-degree, reflecting a distinct role as an exceptionally active recipient who narrated from many teachers; the divergence between the in-degree and out-degree rankings confirms that dominant sources of narration are not necessarily the most active receivers.

The eigenvector scores drop sharply after the top four narrators (from 0.2659 for Ibnu Abbas to 0.1723 for Anas), indicating that influence is concentrated in a small, tightly interconnected core. Closeness scores among the top ten are very high and closely spaced (0.9630–1.0000) because, within the strongest-weight subgraph, central narrators reach others through very short paths. In betweenness, the recurrence of Abu Hurairah, Syu'bah, and Sufyan among the top ranks signals a dual role as both influential and connective nodes.

E. Historical Interpretation of the Dominance Pattern

The structural dominance recovered by the metrics aligns with the classical biographical record. Nawawi [23] records that Abu Hurairah narrated 5,374 hadith, the highest of any companion, far ahead of Abdullah bin Umar (2,630) and Aisyah (2,210). Two historical factors explain that volume—his full-time attendance on the Prophet as one of the *Ashabu al-Suffah*, unencumbered by worldly occupation, and a memorization capacity that classical reports associate with a specific supplication of the Prophet [23]. Structurally, this manifests as the node with the most receiving students and the broadest connectivity to other influential narrators.

Sufyan's dominance, by contrast, is confined to out-degree—the metric counting distinct teachers as sources—which matches his historical reputation as an outstanding *recipient* rather than a center of instruction. Among the classical assessments compiled by Rokhibi *et al.* [24], Yahya bin Ma'in held that no one knew the hadith of Al-A'mash better than Sufyan al-Thawri, and Ibnu Abi Khaitsamah relayed his father's judgment that Sufyan al-Thawri was Al-A'mash's most *ahfaz* student. His standing was thus built on receiving narration with high precision rather than on being a reference for many students, exactly the asymmetry the directed graph exposes. This identification takes "Sufyan" to be Sufyan al-Thawri as the most prominent bearer of that name in classical transmission, since the dataset does not carry full *nasab* to confirm a single identity.

Both patterns reinforce a principle of hadith scholarship: a narrator's role cannot be reduced to one number, but is understood through position within the chain, as traditionally studied through *rijal* and *tabaqat* literature. Distinguishing relation direction through in-degree and out-degree reproduces quantitatively, across 177,047 narrators, the role distinction that classical scholars mapped qualitatively.

F. Dashboard Implementation

The analytical results were stored in Supabase and served through the React dashboard. Fig. 2 shows the Narrator Relation Graph page, the core of the system, rendering narrators as nodes and teacher–student relations as directed edges on an HTML5 Canvas. A left filter panel provides search and the five centrality metrics; selecting a metric drives node size and color so that the most central narrators appear largest and centered, while an overlay card updates to show the selected metric's formula and the narrator role it measures, addressing the semantic-accessibility requirement. For readability the view displays a representative subset of the top 50 narrators with 1,000 relations. Selecting a node opens a right-hand detail panel with the narrator's connected teacher and student counts, all five centrality values, and the linked narrators, while highlighting the node's direct relations. The graph supports zoom, pan, drag, node selection, and auto-fit reset.

The remaining three pages complement this view. The *Narrator Statistics* page summarizes the network with total-narrator (177,047) and total-relation (776,798) indicator cards and five bar charts of the top ten narrators per metric. The

TABLE VII
FRAME RATE DURING GRAPH INTERACTION

Scenario	Mean FPS	Min FPS
Zoom in/out (scroll wheel)	57.0	51
Pan canvas	61.0	60
Drag one node	60.7	60

Relation Table page lists all 776,798 weighted teacher–student relations across 777 pages with pagination, sorting, and search; by default it is sorted by weight, placing Ibnu Umar–Nafi' (11,346) at the top. The *Centrality Table* page lists all 177,047 narrators with their five centrality scores across 178 pages, also paginated, sortable, and searchable. The dashboard was deployed online and is publicly accessible without installation.

G. Functional Testing and Output Validity

All 14 black-box scenarios produced the expected output, a 100% success rate, so the dashboard required no corrective iteration between design and build. The scenarios fall into three groups: navigation and data loading (1–2), the Narrator Relation Graph page including selection, metric filtering, search, zoom, drag, and reset (3–9), and the statistics and table pages with pagination and sorting (10–14).

Functional testing confirms that a feature displays data, not that the displayed data is sound, so output validity was examined on three additional levels. On *input validity*, name normalization prevents one narrator from splitting across several nodes, while the converse risk—several narrators merging into one node—was quantified at 9.85% of resolution events rather than assumed absent. On *presentation consistency*, the dashboard never recomputes centrality client-side but reads precomputed values from Supabase, and the data-loading scenario verified that displayed values match database contents. On *external validity*, the narrators the system identifies as most central agree with independent manual scholarship, as detailed in Section IV-J.

H. Non-Functional Performance

Frame rate stayed comfortably above the jank threshold in all three interactions (Table VII). Zoom averaged 57.0 FPS with a 51 FPS minimum, slightly below pan (61.0) and node drag (60.7), consistent with zoom recomputing view transform and scale on every scroll-wheel event in addition to carrying the same per-frame physics and render load as the others. The difference is measurable but too small to be perceived as stutter.

Browser memory did not scale proportionally with node count over the range that could be tested, staying between 8.8 and 11.1 MB from $N = 50$ to $N = 500$ (Table VIII). At $N = 1{,}000$—still under 1% of the 177,047 narrators in the network—the request never completed within the observation window. Memory usage at full network scale therefore cannot be measured empirically on the current architecture, because data fetching fails long before that scale is reached. This makes the decision to render a representative 50-node subset a matter of architectural capacity as much as visual readability.

TABLE VIII
BROWSER HEAP USAGE VS. NODE COUNT

N (requested)	Nodes received	Heap usage
50	50	8.8 MB
100	100	11.1 MB
250	250	10.1 MB
500	500	10.3 MB
1,000	Total failure	—

TABLE IX
LOAD TESTING PER FIXED VU STAGE (20 S EACH)

VU	Throughput	P95 latency	Max latency	Failure rate
10	12.6 req/s	595 ms	1.11 s	0%
50	29.1 req/s	2.76 s	7.43 s	0%
100	27.7 req/s	7.63 s	15.29 s	0%

Load testing over a 95 s ramp of $0 \rightarrow 10 \rightarrow 50 \rightarrow 100 \rightarrow 0$ VUs failed the p(95) latency threshold—4.68 s actual against a 1,000 ms target, with a 1.31 s mean and a 12.72 s maximum—while passing the failure-rate threshold with 0% failures (2,300 of 2,300 checks) at an average throughput of 23.9 requests/s. The fixed-stage breakdown in Table IX locates the limit: the system meets the 1 s target only at 10 concurrent users (p95 595 ms); at 50 VUs p95 has already reached 2.76 s. At 100 VUs throughput actually falls from 29.1 to 27.7 requests/s while p95 climbs to 7.63 s, the signature of saturation—requests queue rather than being rejected, which is why the failure rate remains 0% instead of returning 429 codes. The practical breaking point therefore lies between 10 and 50 concurrent users on the Supabase tier used. These three tests were run with shortened repetition counts for initial verification, so the figures are valid preliminary measurements that warrant larger-sample repetition before being treated as final.

I. Usability Evaluation

All 18 respondents were aged 20–25, and only one (5.6%) had an academic or professional background in hadith studies, so the results reflect general-user perception rather than the primary target audience and should be read as a preliminary indication. Ten respondents (55.6%) rarely used comparable graph visualization applications and four (22.2%) never had.

Item means were highest for feature integration (4.33) and ease of use (4.22), and lowest for the negatively worded inconsistency item (2.06)—a positive result, since low agreement there means respondents did not perceive inconsistency. The adapted SUS statistics appear in Table X: a mean of 71.23 on a 0–100 scale, above the scale midpoint, with substantial spread (SD 15.85, range 35.71–92.86). Given the seven-item instrument and the respondent profile, this score is best read as

TABLE X
ADAPTED SUS SCORES (7 ITEMS, $n = 18$)

Statistic	Value
Respondents (n)	18
Mean	71.23
Median	73.22
Standard deviation	15.85
Minimum	35.71
Maximum	92.86

TABLE XI
COMPARISON WITH PRIOR SANAD NETWORK STUDIES

Study	KG	SNA	Web	Public	Perf.
Farooqi *et al.* [5]	Yes	Partial	No	No	No
Shuaib *et al.* [9]	No	No	No	No	No
Mahmoud *et al.* [14]	Yes	No	No	No	No
Ahmad *et al.* [18]	Yes	Yes	No	No	No
KASHAF [19]	Yes	No	Yes	Partial	No
This work	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

an early positive signal rather than a standard, target-audience-representative measure.

Open-ended responses named the interface design, the ease of tracing teacher–student relations through the graph, and the search and click-through features as what respondents liked most. The most frequent obstacles were the absence of onboarding guidance and unfamiliarity with what the centrality metrics mean, followed by label crowding when the graph is dense—and the suggestions matched: add metric explanations or tooltips, and reduce label overlap.

J. Comparison with Prior Work

Table XI positions this work against prior *sanad* network studies. Earlier work successfully models and analyzes narrator networks at scale, but the outputs are typically static and tied to desktop software such as Gephi and Cytoscape, restricting access to expert users [5], [9]. The principal advantage of this work is the presentation of the analysis as an interactive, publicly accessible web dashboard that integrates a large-scale narrator KG with all five centrality metrics and exploration features in a single interface.

A direct numerical comparison of computational efficiency against Neo4j [5] or NarratorsKG [14] is not possible, because both publications focus on dataset construction and disambiguation accuracy—Farooqi *et al.* report a graph of 2,092 nodes and 77,797 edges, and Mahmoud *et al.* report 95.2% accuracy with a 23.2% *sanad* error rate—without measuring processing time, query latency, or any other efficiency metric for the graph systems they build. This work supplies those missing figures at a scale of 177,047 narrators: about 13 s to build the full graph and compute the degree- and eigenvector-based metrics, and a 595 ms P95 API latency at 10 concurrent users.

The centrality results also agree with independent manual scholarship. Mufid and Sattar [7], applying Motzki's *isnad-cum-matn* method to a single hadith, found that Abu Hurairah built a far wider *isnad* bundle than Abu Sa'id al-Khudri. That small-scale qualitative finding is consistent with the large-scale quantitative result here, which places Abu Hurairah first in four of five metrics across 177,047 narrators. A similar convergence holds for Ibnu Syihab Az-Zuhri, described by Shuaib *et al.* [9] as the first systematic collector and disseminator of hadith: in this network his relation with Ma'mar carries the seventh-highest weight overall (4,381), so his central position is detected structurally as well.

Finally, this work carries out the analysis that Farooqi *et al.* [5] explicitly identified as applicable to their dataset but left undone, executing it directly at full scale on 177,047 narrators.

K. Limitations

Closeness and betweenness centrality were computed on the strongest-weight subgraph rather than the full graph due to computational cost; the dataset is static and not automatically updated; and the analysis addresses network structure without assessing *sanad* authenticity from the perspective of hadith science. On the implementation side, the current data-fetching architecture cannot load a graph approaching full scale, so the display remains limited to a representative subset. The usability evaluation is constrained on two counts: only 1 of 18 respondents came from a hadith-studies background, and the questionnaire adapts seven of the ten standard SUS items, so its score cannot be compared against standard 10-item interpretation thresholds. Name validity was likewise not verified exhaustively, so some entries remain generic terms rather than genuine narrator names.

V. CONCLUSION

This paper analyzed the Islamic scholarly network of the hadith *sanad* using a Knowledge Graph with Social Network Analysis and presented the results through an interactive web dashboard. From the Sanadset 650K corpus, 487,451 cleaned *sanad* chains were modeled as a weighted directed graph of 177,047 narrator nodes and 776,798 narration edges; the residual risk of identity merging from kinship resolution was quantified at 9.85% of resolution events rather than left as an untested assumption.

The weighted directed representation proved effective at exposing key narrator roles that conventional text search cannot reach. Text search yields one occurrence count, whereas the five centrality metrics decompose that single number into distinct structural roles: Abu Hurairah ranks first in four of five metrics, while Sufyan and Syu'bah stand out specifically in out-degree and betweenness—a role distinction impossible to recover from frequency alone, and one that agrees with independent manual scholarship.

The results are served through a React and Supabase dashboard that is publicly accessible without installation.

Validation spanned three complementary aspects: 14 black-box scenarios at a 100% success rate, non-functional testing showing adequate graph rendering performance (57–61 FPS) and API resilience with a practical breaking point between 10 and 50 concurrent users, and an adapted SUS evaluation with 18 respondents averaging 71.23.

Future work should compute closeness and betweenness on the full graph using approximation algorithms or parallel computation; incorporate periodic data updates and additional *sanad* sources; disambiguate bare-anchor nodes against biographical (*rijal*/*tabaqat*) sources and complete the prepared 150-row gold-standard review to obtain citable precision and recall; and repeat the usability evaluation with hadith students and researchers using the full 10-item SUS instrument plus task-based testing. The full-scale graph built here also opens the way to Graph Neural Network approaches for link prediction, estimating missing narrators in broken (*mursal*/*munqati'*) chains, extending the text-based continuity verification of Hendawi *et al.* [15] to a structural setting.

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